

In[1306]:=

```
ClearAll["Global`*"];
清除全部

SeedRandom[];
随机种子

(* ===== *)
(*1. 并发暴胀引擎 (完全原生 Union 去重, 无人工排序干预)*)
并集

(* ===== *)
StepEvolution[g_Graph, window_] := Module[
图 模块
{allEdges, activeEdges, nActive, usedEdges, updates, tempE1, tempE2, x, y, z, wCounter,
newVertices, newActive, newFrozen, remainingEdges}, allEdges = EdgeList[g];
边的列表

activeEdges = RandomSample[Cases[allEdges, _UndirectedEdge]];
伪随机采样 模式匹配 无向边

nActive = Length[activeEdges];
长度

If[nActive < 2, Return[{g, 0}]];
如果 返回

usedEdges = {}; updates = {};
Do[tempE1 = activeEdges[[i]];
Do循环

If[! MemberQ[usedEdges, tempE1], Do[tempE2 = activeEdges[[j]];
如果 成员判定 Do循环

If[! MemberQ[usedEdges, tempE2] &&
如果 成员判定

Length[Intersection[List @@ tempE1, List @@ tempE2]] == 1,
长度 交集 列表 列表

AppendTo[updates, {tempE1, tempE2}];
附加

AppendTo[usedEdges, tempE1]; AppendTo[usedEdges, tempE2];
附加 附加

Break[];], {j, i + 1, Min[i + window, nActive]}], {i, 1, nActive}];
跳出循环 最小值

If[Length[updates] == 0, Return[{g, 0}]];
返回 长度 返回

wCounter = Max[VertexList[g]];
顶点列表

newVertices = {}; newActive = {}; newFrozen = {};
Do[y = Intersection[List @@ pair[[1]], List @@ pair[[2]] [[1]];
Do循环 交集 列表 列表

x = Complement[List @@ pair[[1]], {y}] [[1]];
补集 列表

z = Complement[List @@ pair[[2]], {y}] [[1]];
补集 列表

wCounter++;
AppendTo[newVertices, wCounter];
附加

newActive = Join[newActive, {UndirectedEdge[x, z],
连接 无向边

UndirectedEdge[x, wCounter], UndirectedEdge[wCounter, z]}];
无向边 无向边
```

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newFrozen = Join[newFrozen, {DirectedEdge[x, y], DirectedEdge[y, z]}];
      连接      有向边      有向边

{pair, updates}];
remainingEdges = Complement[allEdges, Flatten[updates]];
      补集      压平

{Graph[Union[Join[VertexList[g], newVertices]],
  图      并集      连接      顶点列表
    Union[Join[remainingEdges, newActive, newFrozen]]], Length[updates]]];
      并集      连接      长度

(* =====*)
(*2. 类空超曲面快照采集 (Spacelike Hypersurface Snapshots)*)
(* =====*)
Print[Style[
  打印      样式
    "[1/3] Is currently igniting and expanding the universe, and collecting genuine
      'present moments' of pure space slices from the past...", Blue, Bold, 14]];
      蓝色      粗体

universe = CompleteGraph[6];
      完全图

steps = 80;
snapshots = {};

Monitor[Do[{universe, concurrentCount} = StepEvolution[universe, 50];
  监控      Do循环
    If[concurrentCount > 0, allEdges = EdgeList[universe];
      如果      边的列表
        activeEdges = Cases[allEdges, _UndirectedEdge];
          模式匹配      无向边
        frozenEdges = Cases[allEdges, _DirectedEdge];
          模式匹配      有向边
        (*空间体积 v: 这一刻的活跃节点集合*)
        activeNodes = Union[Flatten[List @@@ activeEdges]];
          并集      压平      列表
        If[Length[activeNodes] > 1, vol = Length[activeNodes];
          ...      长度      长度
          (*物理质量分离 (Fix 1): 只计算这些活跃节点背负的冻结历史沉淀*)
          frozenGraph = Graph[VertexList[universe], frozenEdges];
            图      顶点列表
          rho = Mean[VertexDegree[frozenGraph, #] & /@ activeNodes];
            ...      顶点度
          AppendTo[snapshots, {vol, rho}];];];, {i, 1, steps}],
      附加

    ProgressIndicator[i, {0, steps}]];
      进度指示器

(* =====*)
(*3. 坐标系翻转与内禀相变积分 (Cosmological Lookback Integration)*)
(* =====*)
Print[
  打印

```

□ □ □ □

```

"[2/3]Coordinate system inversion: Take the latest slice as the present (r = 0),
    perform a causal backward journey towards the Big Bang and cut the dark halo..."];
(*快照是正向时间的, 观测者向过去看, 必须反转数组*)
historySlices = Reverse[snapshots];
maxR = Length[historySlices] - 1;

profile = Table[{r, historySlices[[r + 1, 1]], historySlices[[r + 1, 2]]}, {r, 0, maxR}];

densities = profile[[All, 3]];

peakIdx = First[Ordering[densities, -1]]; (*寻找历史长河中冻结密度的最高峰作为相变边界*)

Rcore = profile[[peakIdx, 1]];

coreMass = 0;
darkMass = 0;

For[i = 1, i ≤ Length[profile], i++, r = profile[[i, 1]];
    vol = profile[[i, 2]];
    rho = profile[[i, 3]];
    If[r ≤ Rcore, coreMass += vol * rho, darkMass += vol * rho];];

tdmRatio = darkMass / Max[1, coreMass];

Print[Style[
    "\n===== ", Gray, Bold]];

Print[Style["📄 [First principle:
    Snapshot calculation results of pure class-empty hypersurfaces]",
    Blue, Bold, 14]];

Print[StringForm[
    "► 宇宙学回溯时间 (Lookback Steps) : r=0 (Now) 到 r=` (Big Bang)", maxR]];

Print[StringForm["► 拓扑相变边界 (Core-Halo Boundary)
    : r=` (Freezing of the peak of sedimentation)", Rcore]];

Print[StringForm["► Core area effective mass (recent spatial accumulation) : `",
    Round[coreMass, 0.1]]];

```

```

Print[
|打印
StringForm["► Causal relics of dark matter (ancient historical accumulation) : ``",
|字符串形式
Round[darkMass, 0.1]]];
|舍入

Print[
|打印
Style["=====", Gray, Bold]];
|样式 |灰色 |粗体

Print[Style[StringForm["\n☯ Topological dark mass vs. Core visible mass ≈ `` : 1",
|打印 |样式 |字符串形式
Round[tdmRatio, 0.4]], Darker[Green], Bold, 18]];
|舍入 |暗黑化 |绿色 |粗体

Print[Style[
|打印 |样式
"(Strictly separating: Only using the 'active node space volume' to weight the
'frozen historical quality density',
completely eliminating geometric artifacts)", Gray, 12]];
|灰色

(* =====*)
(*4. 绘制跨越宇宙红移的内禀演化图*)
(* =====*)
Print["\n[3/3] We are
|打印
currently drawing the evolution picture of dark matter on the cosmic timeline..."];

Print[ListLinePlot[profile[[All, {1, 3}]], PlotStyle → Directive[Purple, Thick], Epilog →
|打印 |绘制点集的线条 |全部 |绘图样式 |指令 |紫色 |粗 |绘制主图后绘制
{Directive[Red, Dashed, Thick], Line[{Rcore, 0}, {Rcore, Max[densities] * 1.1}]},
|指令 |红色 |虚线 |粗 |线段 |最大值
Text[Style["It can be seen that the core area (in the recent period)",
|文本 |样式
12, Darker[Red], Background → White], {Rcore / 2, Max[densities] * 0.9}],
|暗黑化 |红色 |背景色 |白色 |最大值
Text[Style["Causal Shadow Zone (Ancient)", 12, Purple, Background → White],
|文本 |样式 |紫色 |背景色 |白色
{Rcore + (maxR - Rcore) / 2, Max[densities] * 0.9}]],
|最大值

PlotRange → {0, Max[densities] * 1.2}, PlotLabel →
|绘制范围 |最大值 |绘图标签
Style["Cosmological hypersurface density: Space history accumulation vs.
|样式
Recombination time", Bold, 14], AxesLabel → {"回溯时间 r (Lookback Steps)",
|粗体 |坐标轴标签
"Average freezing degree (effective quality) of single slice"},
ImageSize → Large, GridLines → Automatic]];
|图像尺寸 |大 |网格线 |自动

[1/3] Is currently igniting and expanding the universe, and collecting
genuine 'present moments' of pure space slices from the past....

```

[2/3]Coordinate system inversion: Take the latest slice as the present ($r = 0$),
perform a causal backward journey towards the Big Bang and cut the dark halo...

=====

[First principle: Snapshot calculation results of pure class-empty hypersurfaces]

- ▶ 宇宙学回溯时间 (Lookback Steps) : $r=0$ (Now) 到 $r=79$ (Big Bang)
- ▶ 拓扑相变边界 (Core-Halo Boundary) : $r=44$ (Freezing of the peak of sedimentation)
- ▶ Core area effective mass (recent spatial accumulation) : 805555.``
- ▶ Causal relics of dark matter (ancient historical accumulation) : 130394.``

=====

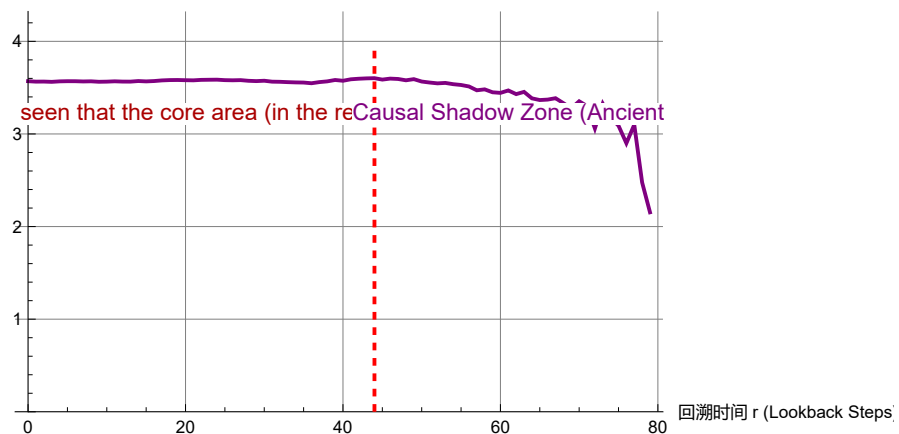
Topological dark mass vs. Core visible mass $\approx 0.`` : 1$

(Strictly separating: Only using the 'active node space volume' to weight the
'frozen historical quality density', completely eliminating geometric artifacts)

[3/3] We are currently drawing the evolution picture of dark matter on the cosmic timeline...

cosmological hypersurface density: Space history accumulation vs. Recombination tim

Average freezing degree (effective quality) of single slice



In[1336]:=

Print[Style[

"\n=== [Strict Correction] Dynamic BA Zero Model Class Empty Snapshot Blind Test



===", Purple, Bold, 16]];





(*确保 TCCT 的动态快照数据还在*)

If[Not[ValueQ[snapshots]] || Not[ValueQ[profile]],

Print["[Error] Unable to locate TCCT snapshot.



Please rerun your concurrent engine code!";

Abort[]];



```

nodeN = VertexCount[universe];
|顶点数
edgeM = Max[1, Round[EdgeCount[universe] / nodeN]];
|最大值 |舍入 |边数
steps = Length[snapshots];
|长度

Print[
|打印
  StringForm["[1/3] Generate the BA model using strict dynamic snapshots... Steps:
|字符串形式
    `` , Noise level m: ``", steps, edgeM]];

baSnapshots = {};
baGraph = Graph[Range[edgeM + 1],
|图 |范围
  UndirectedEdge @@@ Partition[Range[edgeM + 1], 2, 1, 1]]; (*初始小图*)
|无向边 |划分 |范围

Monitor[Do[(*BA 演化: 每步加入1个新节点, 连出 m 条边*) newNode = VertexCount[baGraph] + 1;
|监控 |Do循环 |顶点数
  existingNodes = VertexList[baGraph];
|顶点列表
  degrees = VertexDegree[baGraph];
|顶点度
  probs = degrees / Total[degrees];
|总计
  targets = RandomSample[existingNodes → probs, Min[edgeM, Length[existingNodes]]];
|伪随机采样 |最小值 |长度
  newEdges = UndirectedEdge[newNode, #] & /@ targets;
|无向边
  baGraph =
    Graph[Append[VertexList[baGraph], newNode], Join[EdgeList[baGraph], newEdges]];
|图 |追加 |顶点列表 |连接 |边的列表
  (*绝对公平的当场快照: 当前表面的体积和连通度*) volBA = 1;
  (*每次新增的当下空间节点数为 1*) rhoBA = Length[targets];
|长度
  (*延伸向历史的因果沉淀边数*) AppendTo[baSnapshots, {volBA, rhoBA}];, {i, 1, steps}],
|附加

  ProgressIndicator[i, {0, steps}]];
|进度指示器

Print["[2/3]Generate the BA model using strict dynamic snapshots... Steps:
|打印
  `` , Noise level m: `` Reverse the cosmological time axis,
|反向排序
  redraw the intrinsic evolution diagram..."];
historyBA = Reverse[baSnapshots];
|反向排序
maxR = Length[historyBA] - 1;
|长度

baProfileDyn = Table[{r, historyBA[[r + 1, 1]], historyBA[[r + 1, 2]]}, {r, 0, maxR}];
|表格

```

```
Print["\n[3/3]Generating an absolutely fair comparison chart..."];
```

打印

```
plotTCCT = ListLinePlot[profile[[All, {1, 3}]], PlotStyle → Directive[Purple, Thick],
  绘制点集的线条 全部 绘图样式 指令 紫色 粗
```

```
PlotLegends → {"TCCT (Spontaneous Phase Transition: Halo and Core)"}];
  绘图的图例
```

```
plotBADyn = ListLinePlot[baProfileDyn[[All, {1, 3}]],
  绘制点集的线条 全部
```

```
PlotStyle → Directive[Gray, Dashed, Thick], PlotLegends → {"BA"}];
  绘图样式 指令 灰色 虚线 粗 绘图的图例
```

```
Print[Show[{plotTCCT, plotBADyn}, PlotRange → {0, Max[profile[[All, 3]] * 1.5},
  打印 显示 绘制范围 最大值 全部
```

```
PlotLabel → Style["Strictly dynamic hypersurface comparison:
  绘图标签 样式
```

```
spontaneous emergence vs stagnant background noise",
  Bold, 14], AxesLabel → {"Backward time r (Lookback Steps)",
  粗体 坐标轴标签 后退
```

```
"Average historical accumulation (effective mass) of single slice",
  ImageSize → Large, GridLines → Automatic]];
  图像尺寸 大 网格线 自动
```

=== [Strict Correction] Dynamic BA

Zero Model Class Empty Snapshot Blind Test ===

```
[1/3] Generate the BA model using strict dynamic snapshots... Steps: 80, Noise level m: 3
```

```
[2/3]Generate the BA model using strict dynamic snapshots... Steps: `` , Noise level m:
  `` Reverse the cosmological time axis, redraw the intrinsic evolution diagram...
```

```
[3/3]Generating an absolutely fair comparison chart...
```

dynamic hypersurface comparison: spontaneous emergence vs stagnant background
Average historical accumulation (effective mass) of single slice

